

Homework1

Advanced Statistics and Data Analysis

Davide Risso

Sophie Grace Parolin

Health policy in the United States is complicated, and several forms of healthcare coverage exist, including both coverage by federal government-led healthcare policy, and by private insurance companies. Before making any inference about the relationship between health condition and health policy, it is important for us to have a general idea about healthcare economics in the United States. Thus, we are interested in getting sense of healthcare coverage and healthcare spending across States. More specifically, the questions are:

1. Is there a relationship between healthcare coverage and healthcare spending in the United States?
2. How does the spending distribution change across geographic regions in the United States?
3. Does the relationship between healthcare coverage and healthcare spending in the United States change from 2013 to 2014?

Guided solution

1. Read-in the data: use the `c` function (`readr` package) to read the `healthcare-coverage.csv` and the `healthcare-spending.csv` files (found [here](#) at `data/KFF`).

It looks like the first two lines are descriptive and not useful. We will instruct R to skip them using the `skip` argument in `read_csv()`. The third line appears to contain the column names, and the actual data starts on the fourth line. Additionally, there are a lot of NAs at the end of the data frame because most of that section is not useful.

```
coverage <- read_csv("healthcare-coverage.csv", skip = 2, col_names = TRUE)
coverage <- read_csv("healthcare-coverage.csv", skip = 2, col_names = TRUE,
                     n_max = which(coverage$Location == "Notes")-1)

spending <- read_csv("healthcare-spending.csv", skip = 2, col_names = TRUE)
```

```

spending <- read_csv("healthcare-spending.csv", skip = 2, col_names = TRUE,
                     n_max = which(spending$Location == "Notes")-1)

```

Let's see which variables are included in the datasets.

```
colnames(coverage)
```

```

[1] "Location"          "2013__Employer"    "2013__Non-Group"
[4] "2013__Medicaid"   "2013__Medicare"    "2013__Other Public"
[7] "2013__Uninsured"   "2013__Total"       "2014__Employer"
[10] "2014__Non-Group"   "2014__Medicaid"    "2014__Medicare"
[13] "2014__Other Public" "2014__Uninsured"    "2014__Total"
[16] "2015__Employer"    "2015__Non-Group"    "2015__Medicaid"
[19] "2015__Medicare"    "2015__Other Public" "2015__Uninsured"
[22] "2015__Total"       "2016__Employer"    "2016__Non-Group"
[25] "2016__Medicaid"   "2016__Medicare"    "2016__Other Public"
[28] "2016__Uninsured"   "2016__Total"

```

```
colnames(spending)
```

```

[1] "Location"          "1991__Total Health Spending"
[3] "1992__Total Health Spending" "1993__Total Health Spending"
[5] "1994__Total Health Spending" "1995__Total Health Spending"
[7] "1996__Total Health Spending" "1997__Total Health Spending"
[9] "1998__Total Health Spending" "1999__Total Health Spending"
[11] "2000__Total Health Spending" "2001__Total Health Spending"
[13] "2002__Total Health Spending" "2003__Total Health Spending"
[15] "2004__Total Health Spending" "2005__Total Health Spending"
[17] "2006__Total Health Spending" "2007__Total Health Spending"
[19] "2008__Total Health Spending" "2009__Total Health Spending"
[21] "2010__Total Health Spending" "2011__Total Health Spending"
[23] "2012__Total Health Spending" "2013__Total Health Spending"
[25] "2014__Total Health Spending"

```

Let's limit the datasets to the years 2013 and 2014.

```

coverage <- coverage |>
  select (Location, contains("2013"), contains("2014"))
spending <- spending[,c("Location", "2013__Total Health Spending",
                       "2014__Total Health Spending")]

```

```
head(coverage)
```

```
# A tibble: 6 x 15
  Location `2013__Employer` `2013__Non-Group` `2013__Medicaid` `2013__Medicare`
  <chr>          <dbl>          <dbl>          <dbl>          <dbl>
1 United S~      155696900      13816000      54919100      40876300
2 Alabama        2126500        174200        869700        783000
3 Alaska         364900         24000         95000         55200
4 Arizona        2883800        170800        1346100        842000
5 Arkansas       1128800        155600        600800        515200
6 Californ~     17747300      1986400      8344800      3828500
# i 10 more variables: `2013__Other Public` <chr>, `2013__Uninsured` <dbl>,
#   `2013__Total` <dbl>, `2014__Employer` <dbl>, `2014__Non-Group` <dbl>,
#   `2014__Medicaid` <dbl>, `2014__Medicare` <dbl>, `2014__Other Public` <chr>,
#   `2014__Uninsured` <dbl>, `2014__Total` <dbl>
```

```
head(spending)
```

```
# A tibble: 6 x 3
  Location `2013__Total Health Spending` `2014__Total Health Spending`
  <chr>          <dbl>          <dbl>
1 United States 2435624      2562824
2 Alabama      33788       35263
3 Alaska       7684       8151
4 Arizona     41481      43356
5 Arkansas    20500      21980
6 California  278168     291989
```

```
glimpse(coverage)
```

```
Rows: 52
```

```
Columns: 15
```

```
$ Location      <chr> "United States", "Alabama", "Alaska", "Arizona", ~
$ `2013__Employer` <dbl> 155696900, 2126500, 364900, 2883800, 1128800, 177~
$ `2013__Non-Group` <dbl> 13816000, 174200, 24000, 170800, 155600, 1986400,~
$ `2013__Medicaid` <dbl> 54919100, 869700, 95000, 1346100, 600800, 8344800~
$ `2013__Medicare` <dbl> 40876300, 783000, 55200, 842000, 515200, 3828500,~
$ `2013__Other Public` <chr> "6295400", "85600", "60600", "N/A", "67600", "675~
$ `2013__Uninsured` <dbl> 41795100, 724800, 102200, 1223000, 436800, 559410~
$ `2013__Total` <dbl> 313401200, 4763900, 702000, 6603100, 2904800, 381~
```

```

$ `2014__Employer`      <dbl> 154347500, 2202800, 345300, 2835200, 1176500, 177~
$ `2014__Non-Group`     <dbl> 19313000, 288900, 26800, 333500, 231700, 2778800,~
$ `2014__Medicaid`     <dbl> 61650400, 891900, 130100, 1639400, 639200, 961880~
$ `2014__Medicare`      <dbl> 41896500, 718400, 55300, 911100, 479400, 4049000,~
$ `2014__Other Public`  <chr> "5985000", "143900", "37300", "N/A", "82000", "63~
$ `2014__Uninsured`     <dbl> 32967500, 522200, 100800, 827100, 287200, 3916700~
$ `2014__Total`         <dbl> 316159900, 4768000, 695700, 6657200, 2896000, 387~

```

```
glimpse(spending)
```

```

Rows: 52
Columns: 3
$ Location                <chr> "United States", "Alabama", "Alaska", "A~
$ `2013__Total Health Spending` <dbl> 2435624, 33788, 7684, 41481, 20500, 2781~
$ `2014__Total Health Spending` <dbl> 2562824, 35263, 8151, 43356, 21980, 2919~

```

The variables 2013__Other Public and 2014__Other Public are stored as characters (chr), so we convert them to numeric.

```

coverage$`2014__Other Public` = as.numeric(coverage$`2014__Other Public`)
coverage$`2013__Other Public` = as.numeric(coverage$`2013__Other Public`)

```

We need to transform the data into a “tidy” format. To do this, we convert the data from wide format to long format by creating two new columns: `year_type` and `coverage`. We also keep the `Location` column unchanged, as it contains observational-level data.

```

coverage <- coverage |>
  pivot_longer(cols = -Location, names_to = "year_type",
               values_to = "tot_coverage")
coverage

```

```

# A tibble: 728 x 3
  Location      year_type      tot_coverage
  <chr>         <chr>         <dbl>
1 United States 2013__Employer 155696900
2 United States 2013__Non-Group 13816000
3 United States 2013__Medicaid 54919100
4 United States 2013__Medicare 40876300
5 United States 2013__Other Public 6295400
6 United States 2013__Uninsured 41795100

```

```

7 United States 2013__Total          313401200
8 United States 2014__Employer       154347500
9 United States 2014__Non-Group      19313000
10 United States 2014__Medicaid      61650400
# i 718 more rows

```

Now, each row represents a single observation, consisting of a `Location`, a `year_type`, and `coverage`. It would be helpful to split the `year_type` column into two separate columns: `year` and `type` (representing the health coverage type).

```

coverage <-
  coverage |>
  separate(year_type, sep="__",
            into=c("year", "type"),
            convert = TRUE)
# convert = TRUE --> argument to convert the character to an integer.
coverage

```

```

# A tibble: 728 x 4
  Location      year type      tot_coverage
  <chr>      <int> <chr>      <dbl>
1 United States  2013 Employer    155696900
2 United States  2013 Non-Group   13816000
3 United States  2013 Medicaid    54919100
4 United States  2013 Medicare    40876300
5 United States  2013 Other Public  6295400
6 United States  2013 Uninsured   41795100
7 United States  2013 Total      313401200
8 United States  2014 Employer    154347500
9 United States  2014 Non-Group   19313000
10 United States  2014 Medicaid    61650400
# i 718 more rows

```

```
table(coverage$type)
```

Employer	Medicaid	Medicare	Non-Group	Other Public	Total
104	104	104	104	104	104
Uninsured					
104					

Note that the `total` type is not really a formal type of healthcare coverage. It represents just the total number of people in the state. This is a useful information and we can include it as a column called `tot_pop`.

```
pop <- coverage |>
  filter(type == "Total") |>
  select(Location, year, tot_coverage)
head(pop)
```

```
# A tibble: 6 x 3
  Location      year tot_coverage
  <chr>         <int>      <dbl>
1 United States 2013      313401200
2 United States 2014      316159900
3 Alabama       2013       4763900
4 Alabama       2014       4768000
5 Alaska        2013       702000
6 Alaska        2014       695700
```

```
coverage <- coverage |>
  filter(type != "Total") |>
  left_join(pop, by = c("Location", "year")) |>
  rename(tot_coverage = tot_coverage.x, tot_pop = tot_coverage.y)
coverage
```

```
# A tibble: 624 x 5
  Location      year type      tot_coverage tot_pop
  <chr>         <int> <chr>          <dbl>      <dbl>
1 United States 2013 Employer      155696900 313401200
2 United States 2013 Non-Group    13816000 313401200
3 United States 2013 Medicaid     54919100 313401200
4 United States 2013 Medicare     40876300 313401200
5 United States 2013 Other Public   6295400 313401200
6 United States 2013 Uninsured     41795100 313401200
7 United States 2014 Employer     154347500 316159900
8 United States 2014 Non-Group    19313000 316159900
9 United States 2014 Medicaid     61650400 316159900
10 United States 2014 Medicare     41896500 316159900
# i 614 more rows
```

```
table(coverage$type)
```

Employer	Medicaid	Medicare	Non-Group	Other	Public	Uninsured
104	104	104	104		104	104

Now, let's also transform the `spending` dataset into a tidy format.

```
spending <- spending |>
  pivot_longer(cols = -Location, names_to = "year",
               values_to = "tot_spending")
spending
```

```
# A tibble: 104 x 3
  Location      year tot_spending
  <chr>         <chr>         <dbl>
1 United States 2013__Total Health Spending 2435624
2 United States 2014__Total Health Spending 2562824
3 Alabama      2013__Total Health Spending 33788
4 Alabama      2014__Total Health Spending 35263
5 Alaska       2013__Total Health Spending 7684
6 Alaska       2014__Total Health Spending 8151
7 Arizona      2013__Total Health Spending 41481
8 Arizona      2014__Total Health Spending 43356
9 Arkansas     2013__Total Health Spending 20500
10 Arkansas    2014__Total Health Spending 21980
# i 94 more rows
```

```
spending <-
  spending |>
  separate(year, sep="__",
           into=c("year", "name"),
           convert = TRUE) |>
  select(-name)
spending
```

```
# A tibble: 104 x 3
  Location      year tot_spending
  <chr>         <int>         <dbl>
1 United States 2013         2435624
```

```

2 United States 2014 2562824
3 Alabama      2013 33788
4 Alabama      2014 35263
5 Alaska       2013 7684
6 Alaska       2014 8151
7 Arizona      2013 41481
8 Arizona      2014 43356
9 Arkansas     2013 20500
10 Arkansas    2014 21980
# i 94 more rows

```

2. Load the state information found in the state datasets (datasets package). Note that you need to manually add information about the District of Columbia.

Let's analyze what state contains.

```

library(datasets)
data(state)
unique(state.name)

```

```

[1] "Alabama"      "Alaska"      "Arizona"     "Arkansas"
[5] "California"   "Colorado"    "Connecticut" "Delaware"
[9] "Florida"     "Georgia"     "Hawaii"      "Idaho"
[13] "Illinois"    "Indiana"     "Iowa"        "Kansas"
[17] "Kentucky"    "Louisiana"   "Maine"       "Maryland"
[21] "Massachusetts" "Michigan"    "Minnesota"   "Mississippi"
[25] "Missouri"    "Montana"     "Nebraska"    "Nevada"
[29] "New Hampshire" "New Jersey"  "New Mexico"  "New York"
[33] "North Carolina" "North Dakota" "Ohio"        "Oklahoma"
[37] "Oregon"      "Pennsylvania" "Rhode Island" "South Carolina"
[41] "South Dakota" "Tennessee"   "Texas"       "Utah"
[45] "Vermont"     "Virginia"    "Washington"  "West Virginia"
[49] "Wisconsin"   "Wyoming"

```

```

unique(coverage$Location)

```

```

[1] "United States"      "Alabama"      "Alaska"
[4] "Arizona"            "Arkansas"     "California"
[7] "Colorado"           "Connecticut"  "Delaware"
[10] "District of Columbia" "Florida"      "Georgia"
[13] "Hawaii"             "Idaho"        "Illinois"

```

[16] "Indiana"	"Iowa"	"Kansas"
[19] "Kentucky"	"Louisiana"	"Maine"
[22] "Maryland"	"Massachusetts"	"Michigan"
[25] "Minnesota"	"Mississippi"	"Missouri"
[28] "Montana"	"Nebraska"	"Nevada"
[31] "New Hampshire"	"New Jersey"	"New Mexico"
[34] "New York"	"North Carolina"	"North Dakota"
[37] "Ohio"	"Oklahoma"	"Oregon"
[40] "Pennsylvania"	"Rhode Island"	"South Carolina"
[43] "South Dakota"	"Tennessee"	"Texas"
[46] "Utah"	"Vermont"	"Virginia"
[49] "Washington"	"West Virginia"	"Wisconsin"
[52] "Wyoming"		

We see there are more than 50 states because “United States” and “District of Columbia” are both included.

Let’s create a dataframe with all the variables we will need and manually enter the information related to the District of Columbia.

```
states <- data.frame(
  Location = state.name,
  # Note: We name the variable "Location" for consistency
  # with the other datasets to facilitate the join.
  abb = state.abb,
  region = state.region,
  stringsAsFactors = FALSE
)

# Manually add information about the District of Columbia
dc_data <- data.frame(
  Location = "District of Columbia",
  abb = "DC",
  region = "South",
  stringsAsFactors = FALSE
)
states <- rbind(states, dc_data)
head(states)
```

	Location	abb	region
1	Alabama	AL	South
2	Alaska	AK	West

```
3   Arizona  AZ   West
4   Arkansas AR   South
5   California CA   West
6   Colorado CO   West
```

3. Add the abbreviation and the region of each state to the coverage dataset.

```
coverage<- inner_join(coverage, states, by = "Location")
```

4. Join the coverage and spending datasets.

Both share a Location column and a year column.

```
data_all=inner_join(coverage, spending, by = c("Location", "year"))
str(data_all)
```

```
tibble [612 x 8] (S3: tbl_df/tbl/data.frame)
 $ Location      : chr [1:612] "Alabama" "Alabama" "Alabama" "Alabama" ...
 $ year          : int [1:612] 2013 2013 2013 2013 2013 2013 2013 2014 2014 2014 2014 ...
 $ type          : chr [1:612] "Employer" "Non-Group" "Medicaid" "Medicare" ...
 $ tot_coverage: num [1:612] 2126500 174200 869700 783000 85600 ...
 $ tot_pop       : num [1:612] 4763900 4763900 4763900 4763900 4763900 ...
 $ abb           : chr [1:612] "AL" "AL" "AL" "AL" ...
 $ region        : Factor w/ 4 levels "Northeast","South",...: 2 2 2 2 2 2 2 2 2 2 ...
 $ tot_spending: num [1:612] 33788 33788 33788 33788 33788 ...
```

5. Use ggplot to produce a scatterplot of the proportion of coverage by state vs. the spending per capita. Color code the points by region and add state abbreviations as labels. Use facet to stratify the analysis by type of coverage.

Instead of the absolute number of people who are covered (`tot_coverage`), we need to calculate the proportion of people who are coverage in each state, year and type. In addition, the `tot_spending` column is reported in millions (1e6). Therefore, to calculate `spending_capita` we will need to adjust for this scaling factor to report it on the original scale (just dollars) and then divide by `tot_pop`.

```
data_all <- data_all |>
  mutate(
    spending_capita = (tot_spending*1e6) / tot_pop,
    prop_coverage = tot_coverage/tot_pop
  )
```

Now that the dataset is “tidy”, let’s create the required plot from the assignment.

```
# ?ggplot2::facet_grid

library(ggrepel)

data_all |>
  filter(year=="2013") |>
  ggplot(aes(y = prop_coverage, x = spending_capita,
             color = region)) +
  geom_point(size = 1, alpha = 0.7) +
  geom_text_repel(aes(label=abb)) + # Add state abbreviations
  facet_wrap(~ type, scales="free") + # Stratify by type of coverage
  # and allow each facet to have independent axes
  labs(y = "Coverage Proportion",
       x = "Spending per capita ($)",
       color = "Region") +
  theme_minimal()
```

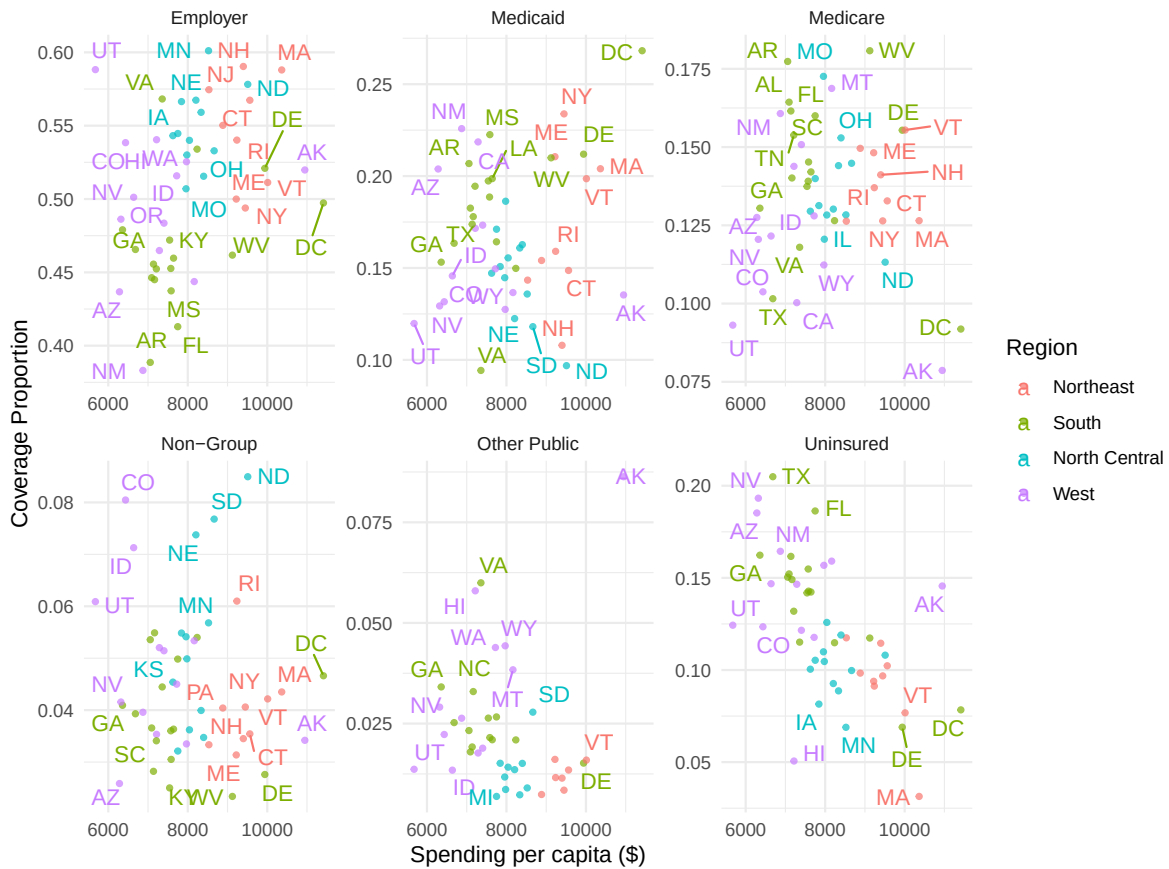


Figure 1: Proportion of Coverage vs. Per Capita Health Spending - 2013

```
data_all |>
  filter(year=="2014") |>
  ggplot(aes(y = prop_coverage, x = spending_capita,
             color = region, label = abb)) +
  geom_point(size = 1, alpha = 0.7) +
  geom_text_repel() +
  facet_wrap(~ type, scales="free") +
  labs(y = "Coverage Proportion",
       x = "Spending per capita ($)",
       color = "Region") +
  theme_minimal()
```

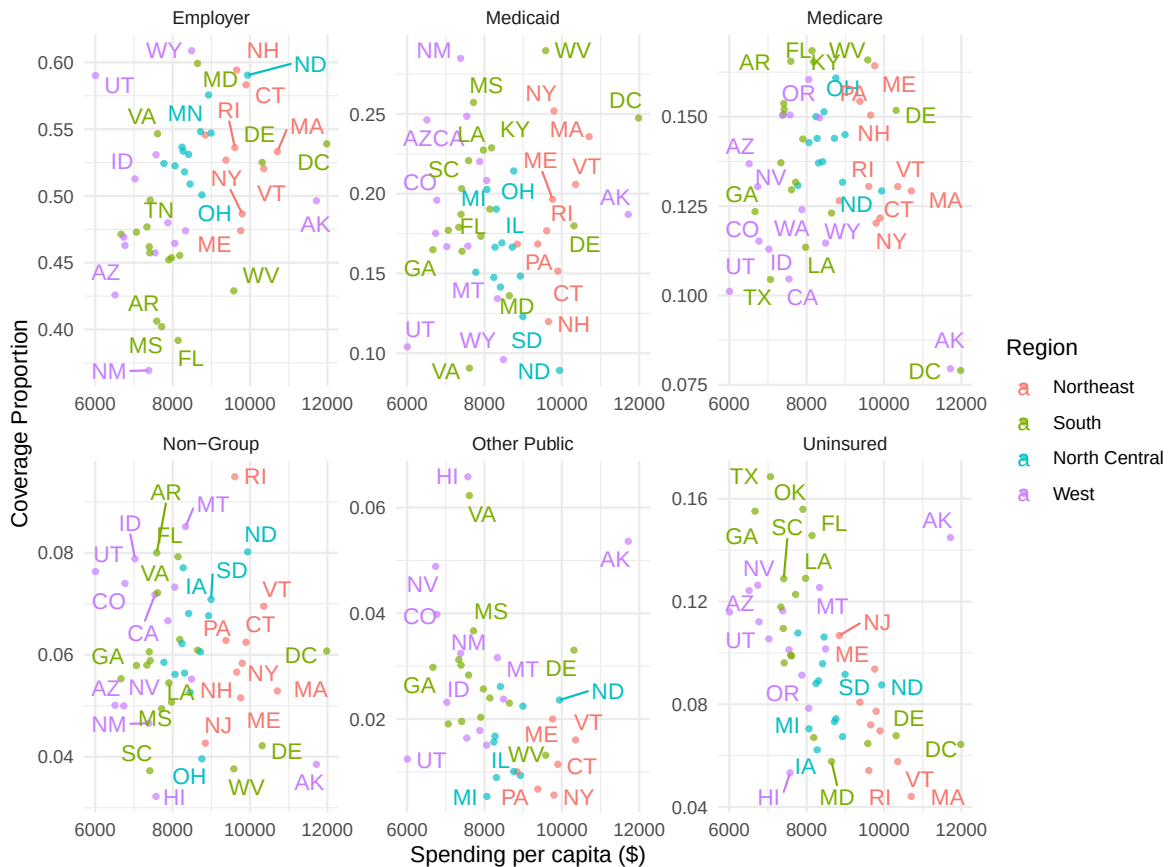


Figure 2: Proportion of Coverage vs. Per Capita Health Spending - 2014

6. Use ggplot to create a boxplot of spending per capita stratified by region.

```
# Create the boxplot
data_all |>
  ggplot(aes(x = region,
             y = spending_capita,
             fill = region)) +
  geom_boxplot(alpha = 0.7) +
  labs(x = "Region",
       y = "Spending per capita ($)") +
  theme_minimal() +
  theme(legend.position = "none")
# Remove legend (colors are already mapped to regions)
```

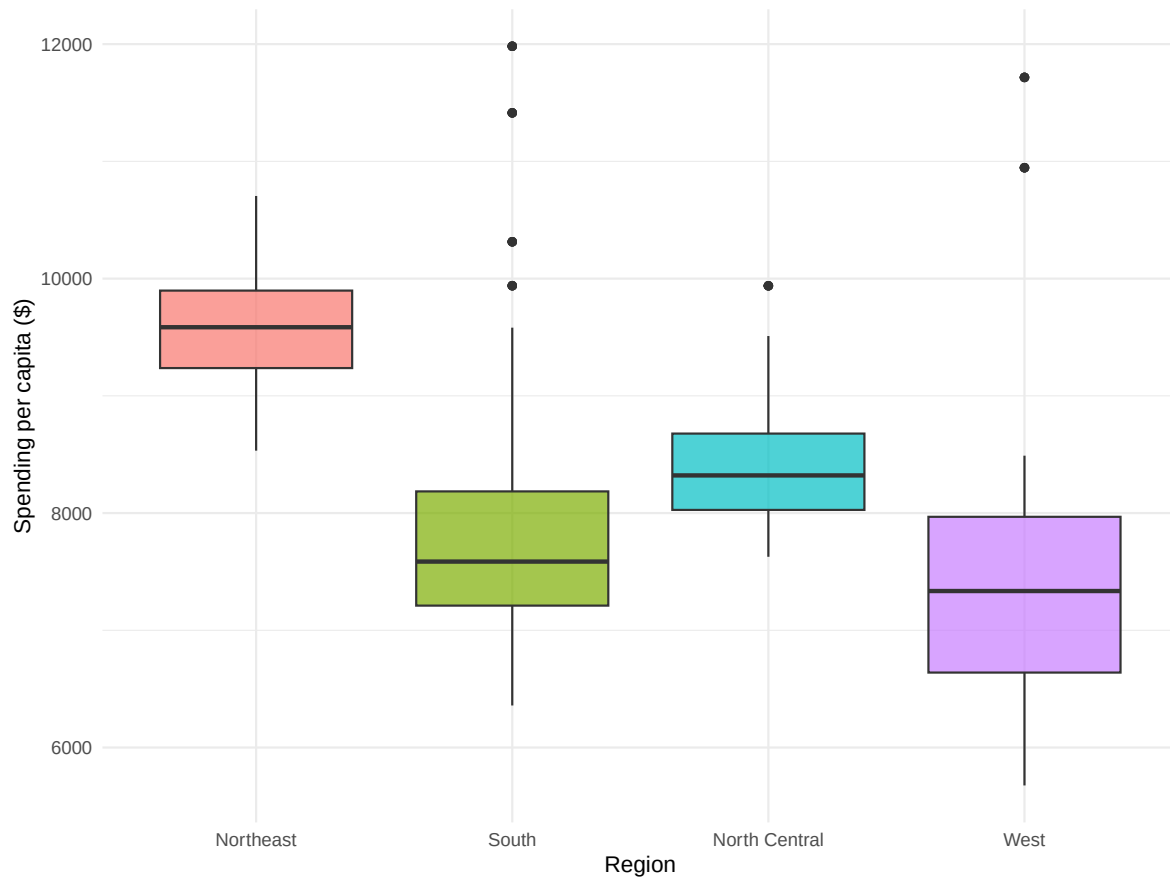


Figure 3: Per Capita Health Spending by Region

- Repeat the graph in point 5 but faceting by year, in addition to coverage type. Hint: use the `facet_grid()` function.

```
data_all |>
  ggplot(aes(y = prop_coverage, x = spending_capita,
             color = region, label = abb)) +
  geom_point(size = 1, alpha = 0.7) +
  geom_text_repel() + # Add state abbreviations
  facet_wrap(type ~ year, scales="free") + # Stratify by year and
  # type of coverage
  labs(y = "Coverage Proportion by State",
       x = "Per Capita Health Spending ($)",
       color = "Region") +
  theme_minimal()
```

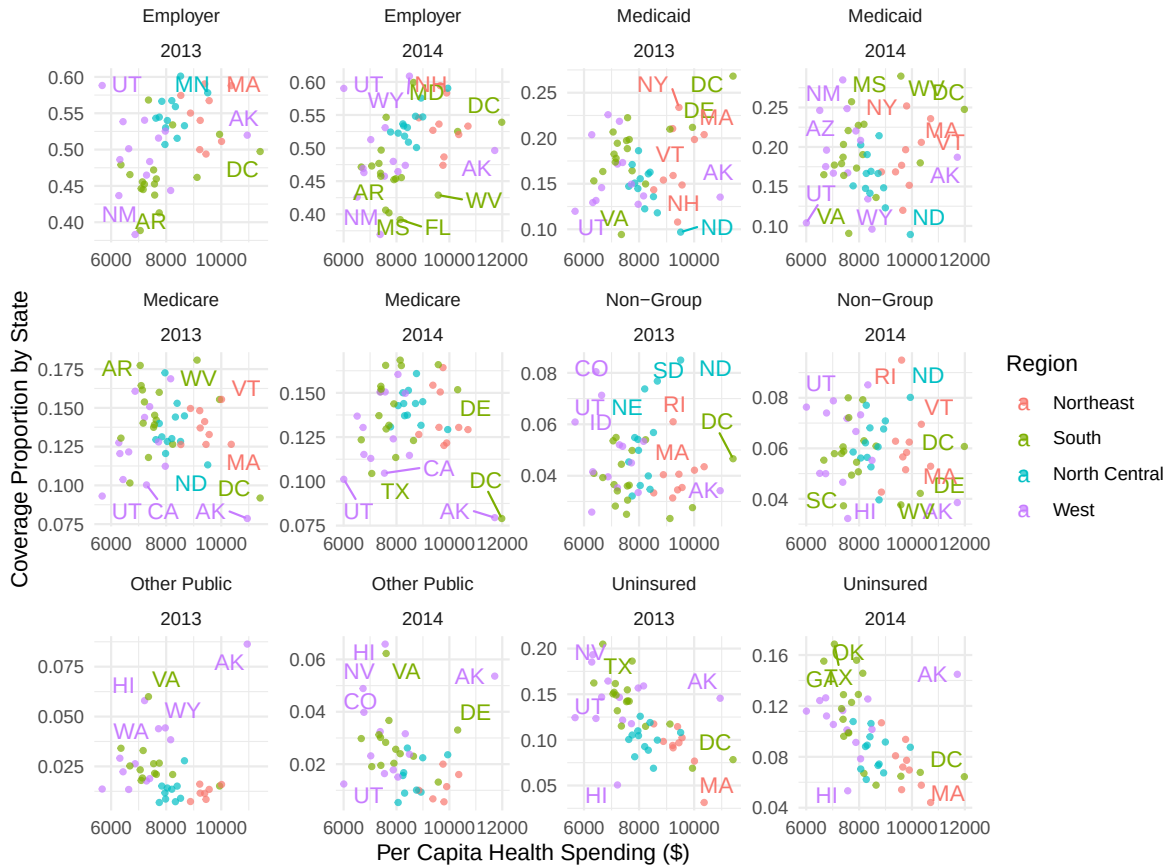


Figure 4: Proportion of Coverage vs. Per Capita Health Spending

Responses to the research questions

1. Is there a relationship between healthcare coverage and healthcare spending in the United States?

In Figure 1 and Figure 2, we observe a positive relationship between healthcare coverage and spending for Employer-based coverage.

Additionally, we find a negative relationship for the Uninsured category: states that spend more tend to have lower proportions of uninsured individuals.

However, for the other coverage types, there is no clear relationship between coverage proportion and per capita healthcare spending. There is no evident trend suggesting that a higher coverage proportion leads to either higher or lower spending.

2. How does the spending distribution change across geographic regions in the United States?

Figure 3 illustrates the per capita health spending distribution across different U.S. regions (Northeast, South, North Central, and West). The figure reveals significant variations in per capita health spending across these regions.

The Northeast has the highest median spending. The South and West regions have the lowest median spending, while the North Central falls in between, exhibiting moderate spending levels with a more compact distribution, indicating relatively uniform spending across states in this region.

Additionally, we observe that the South has several high-spending outliers, notably the District of Columbia, Delaware, and West Virginia, as seen in Figure 1 and Figure 2. The West also has a few outliers, notably Alaska. The North Central region also has an outlier, but it remains closer to the main distribution compared to the other regions.

3. Does the relationship between healthcare coverage and healthcare spending in the United States change from 2013 to 2014?

The facet grid in Figure 4 stratifies the data by year (2013 vs. 2014) and type of coverage, allowing us to compare the relationship between coverage proportion and per capita health spending over time.

The relationship between healthcare coverage proportion and per capita spending appears almost unchanged from 2013 to 2014. While there may be small state-level variations, there is no clear shift in how spending is associated with coverage proportion.

Conclusion

The exploratory analysis through data visualization suggests that higher per capita healthcare spending is positively associated with a higher employer coverage proportion and negatively associated with the proportion of the uninsured population across states.

Note

Please note that there is no single correct way to complete the homework.